(d) Ramachandran plot

(e) Mucopolysaccharide

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06/06/2026 (MORNING)

[This question paper contains 4 printed pages]

Your Roll No. : 06.06.2026 (M)

Sl. No. of Q. Paper : **2063** I

Unique Paper Code : 2232011202

Name of the Paper : Fundamentals of Biomolecules

Name of the Course : **B.Sc.(Hons.) Zoology** (NEP-UGCF)

Semester : II

Time : 2 Hours ***Maximum Marks : 60***

Instructions for Candidates :

(a) Write your Roll No. on the top immediately on receipt of this question paper.

(b) Attempt any **four** questions in **all**. Question **No.1** is compulsory.

(c) Draw a suitable well-labelled diagram wherever necessary.

1. (a) Define the following : 3×1=3

(i) Glycoconjugates

(ii) Isozymes

(iii) Domain

(b) Draw the chemical structures of the following : 4×1=4

(i) Chondroitin-4-sulfate

(ii) Thymine

(iii) Cholesterol

(iv) Histidine

(c) Distinguish between the following : 4×2=8

(i) Anomers and epimers

(ii) Aldoses and ketoses

(iii) Nucleotide and nucleoside

(iv) Fibrous and globular proteins

2. (a) Derive the Michaelis-Menten equation of enzyme kinetics. 9

(b) What are regulatory enzymes? Give their different types with examples. 6

3. (a) Describe Watson and Crick model of double helical structure of DNA using a well-labelled diagram. 9

(b) What are the different forms of DNA? Compare them by outlining their contrasting features. 6

4. (a) What are the various levels of protein structural organization? 9

(b) What are conjugated proteins? Write their types with examples. 6

5. (a) What are structural polysaccharides? Illustrate their structures and explain the key differences among them. 10

(b) What are sphingolipids? What are their different types ? 5

6. Write a short note on any **three** : 5×3=15

(a) Lineweaver-Burk plot

(b) Enzyme inhibition

(c) DNA denaturation